Spatial–Spectral Enhancement Contrastive Learning Network for Few-Shot Hyperspectral Image Classification

Shizhe Zhang, Lina Yang, *Member, IEEE*, Yaojun Cai, Yifeng Tan, *Student Member, IEEE*, Wanxing Zhu, Thomas Xinzhang Wu, *Senior Member, IEEE*, Xichun Li, and Yan Yuan Tang, *Life Fellow, IEEE*

***Abstract*—Few-shot hyperspectral image (HSI) classification plays a crucial role in the field of remote sensing with the rapid development of deep learning. However, few-shot HSI classification remains challenging. When labeled samples are scarce, severe overfitting arises from limited feature generalization ability in effectively capturing and fusing discriminative spatial–spectral features. In this study, a spatial–spectral enhancement contrastive learning network (SSECLN) is proposed for few-shot HSI classification. This network includes three modules: spatial–spectral contrastive learning (SSCL), spatial–spectral cross-attention (SSCA), and partial multikernel convolution (PMC). The SSCL includes multiscale spatial enhancement (MSE), bidirectional spectral enhancement (SSE), and batch contrastive learning (BCL). The SSE extracts spatial structural information at different scales and combines forward and backward spectral context to fully explore spectral dependencies. By introducing same-class sample constraints, BCL not only improves intraclass compactness and interclass separability but also alleviates overfitting problems in few-shot scenarios. The SSCA realizes the dynamic interaction of local spatial information with global spectral information, adaptively accommodating the coupled spatial–spectral features through its inherent mechanism. The PMC is used to extract important spatial–spectral features to improve model efficiency. The results show that the overall accuracy is increased by 5.86%–34.59% (IP), 0.96%–24.28% (KSC), 2.39%–23.40% (Salinas), and 3.86%–35.85% (UP) by using the comparison of SSECLN and the state-of-the-art approaches. Codes are available at https://github.com/Zhangshizhe04/SSECLN**

***Index Terms*—Contrastive learning, few-shot, hyperspectral image (HSI) classification, spatial–spectral feature enhancement.**

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Shizhe Zhang, Lina Yang, Yifeng Tan, and Wanxing Zhu are with the School of Computer, Electronics and Information, Guangxi University, Nanning, Guangxi 530004, China (e-mail: 2413394040@st.gxu.edu.cn; lnyang@gxu.edu.cn; tyf1282331092@163.com; 2313393087@st.gxu.edu.cn).

Yaojun Cai is with Guangxi Key Laboratory of Digital Infrastructure, Nanning, Guangxi 530201, China (e-mail: 646791991@qq.com).

Thomas Xinzhang Wu is with the School of Electrical Engineering, Shanghai Jiaotong University, Shanghai 200240, China (e-mail: wuee@sjtu.edu.cn).

Xichun Li is with the School of Mathematics and Computer Science, Guangxi Normal University for Nationalities, Chongzuo, Guangxi 532200, China (e-mail: lixichun@gxnun.edu.cn).

Yan Yuan Tang is with the Faculty of Science and Technology, University of Macau, Macau, China (e-mail: yytang@umac.mo).

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I. Introduction

With the advancement of Earth observation research, hyperspectral remote sensing technology makes great contributions. Hyperspectral images (HSIs) cover a wider spectral range, including the visible spectrum, near-infrared, mid-infrared, and far-infrared regions, containing dozens or even hundreds of continuous bands [1], [2]. It can identify ground objects that are difficult to distinguish from natural RGB images, making Earth observation more accurate. HSI classification, as a fundamental method in mineral detection [3], precision agriculture [4], and environmental monitoring [5], performs the critical task of predicting the class label of each pixel in HSI.

Traditional machine learning methods such as support vector machine (SVM), random forest (RF), multinomial logistic regression (MLR), and feature selection or extraction methods were used in previous research on HSI classification. These models focus on expert prior knowledge and are difficult to strike a balance between distinguishability and robustness. To address this drawback, deep learning has been introduced into HSI classification. Mou et al. [6] proposed a deep recurrent neural network combined with a multilayer 1-D-CNN to effectively integrate spectral correlations and band variability, capturing deep features in the spectral dimension and thereby improving classification performance. To enhance the utilization of spatial information, Makantasis et al. [7] proposed a 2-D-CNN-based HSI classification method. Spatial features are initially extracted from pixels using the 2-D-CNN, followed by final classification employing a multilayer perceptron (MLP). To simultaneously extract spatial–spectral features, Paoletti et al. [8] proposed 3-D convolutional kernels to directly model the multidimensional structure of hyperspectral data, further enhancing classification accuracy and efficiency. To enhance CNN with attention mechanisms, Dong et al. [9] proposed a network that combines convolutional neural networks and attention, using CNN to reduce dimensionality and noise, and then combining position and channel attention modules to enhance feature expression. To overcome the limitations of attention mechanisms in learning spatial–spectral information from HSIs, Zhang et al. [10] proposed cross-attention to integrate local and global features, with group cascading promoting cross-band information complementarity, and a dual-branch structure enlarging interclass differences.

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However, due to the fact that such methods involve training complex models with massive parameters, their dependence on labeled data is particularly significant. Unfortunately, the collection of pixel-level annotations for HSIs is both laborious and time-consuming. Therefore, it is crucial to develop algorithms that can provide robust performance under few-shot conditions.

Recently, methods related to contrastive loss (CL) [11] have been widely applied in few-shot HSI classification. Contrastive learning is proposed to minimize the distance between similar samples and maximize the distance between samples of different categories. Early research by Xu et al. [12] proposed framework integrates contrastive learning with uncertainty-aware self-training to select reliable pseudo-labels and employs Mahalanobis distance for decision-boundary-aware prototype alignment, thereby improving cross-scene domain adaptation. Chen et al. [13] proposed approach employs contrastive learning on the encoder side integrated with a convolutional block attention module, enhances sample discrimination through data augmentation to generate positive samples, and constructs a supervised CL to improve feature aggregation within classes and separation between classes, thereby optimizing the discriminability of spatial–spectral features.

To improve the quality of data, training samples are transformed into diverse samples through data augmentation strategies. Wu et al. [14] proposed a spatial–spectral hybrid augmentation method, incorporating center-clipped scaling, horizontal flipping, and random permutation of band-selected point values, to facilitate robust feature extraction and improve feature representation and knowledge retention in class-incremental learning. Zhu et al. [15] proposed the approach of generating data augmentation views through random cropping in contrastive learning, generating multiscale patches centered around pixels and making features of different views of the same object similar, thus improving the model's learning of consistency in local features of HSI.

However, existing convolutional neural network methods include what Lee and Kwon [16] proposed: a deeper and broader CNN, which combines multiscale filter banks and residual learning to extract spatial–spectral information. Huang et al. [17] proposed a CNN model that integrates convolution, batch normalization, and an improved triplet loss mechanism to enhance the extraction of spatial–spectral features. However, their method has fixed convolution kernels and convolution positions, making it difficult to extract important information from HSIs more flexibly. To address this limitation, partial multikernel convolution (PMC) has been proposed. Existing cross-attention methods include what Yang et al. [18] proposed: cross-spatial attention generates spatial–spectral features through the correlation between pixels and neighboring pixels, calculates the correlation first, and then normalizes and aggregates the characteristics of neighboring pixels. Peng et al. [19] proposed long-short-term spectral cross-attention to group spectra, with short-term groups capturing adjacent details and long-term groups capturing distant correlations. However, their methods either employ static correlation calculations or grouping strategies, without utilizing dynamic interaction mechanisms, resulting in weak adaptability. To address this limitation, spatial–spectral cross-attention (SSCA) has been proposed. Existing contrastive learning methods include what Li et al. [20] proposed for cross-domain zero-shot HSI classification, a method named MAC-CDZS that adopts a two-stage MLP encoder for feature extraction and integrates a graph convolution decoder with spatial augmentation-based contrastive learning to mitigate class collapse and enhance feature representation. Zhou et al. [21] proposed local contrastive learning (LCL) for the pre-training stage, with encoder and decoder outputs as contrastive pairs. The encoder outputs multiscale features, the decoder for reconstruction, and LCL focuses on central-region features to enhance interbranch mutual learning and central information modeling. However, their method does not use the same-class CL (SCL), which makes the model prone to overfitting local noise from similar samples in the training set. To address this issue, batch contrastive learning (BCL) has been proposed. Existing data augmentation methods include the FSHyper-RGAN strategy proposed by Wei et al. [22], which consists of adversarial generation, data embedding and reconstruction, and relational computation. This method is applied in a few-shot learning framework to enhance feature extraction for HSI classification. Zhang et al. [23] proposed the FPGANDA data augmentation strategy, which retains 60% of the key spectral bands and supplements the rest with GAN-generated bands, balancing the original and generated features. However, their method does not fully consider discriminative spatial–spectral features. To address this situation, multiscale spatial enhancement (MSE) and bidirectional spectral enhancement (SSE) are proposed. In summary, the main contributions are as follows.

1) To extract discriminative spatial–spectral features and suppress similar-sample noise, we propose spatial–spectral contrastive learning (SSCL). It strengthens spatial–spectral feature representation and introduces SCL to alleviate overfitting.
2) To dynamically model the interaction between multiscale spatial and bidirectional spectral information, we propose SSCA. It fuses multiscale spatial and bidirectional spectral features to adaptively capture complex spatial–spectral correlations.
3) To efficiently capture significant spatial–spectral features, we propose PMC. It extracts discriminative features via adaptive convolution kernels and positions to boost model efficiency.

II. Related Work

A. Few-Shot HSI Classification

Few-shot HSI classification aims to recognize ground objects using extremely limited labeled samples, which poses significant challenges due to high dimensionality and overfitting risks. Recent efforts focus on enhancing spatial–spectral feature extraction and leveraging advanced architectures. Liu et al. [24] proposed a spatial–spectral enhancement and fusion network (SSEFN) that adaptively fuses multilevel features to improve discriminability under few-shot conditions. Transformer-based methods, such as the spectral query spatial

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former (SQSFormer) [25], incorporate center-pixel attention to model long-range dependencies, yet they often rely on static attention computations that limit adaptability.

Contrastive learning has been integrated to enforce intraclass compactness. Xi et al. [26] developed a mixed CNN-Transformer with graph contrastive learning (MCTGCL) for Mars HSI classification, demonstrating improved feature separability. Huang et al. [27] combined supervised and contrastive learning in a masked Transformer (SC-SS-MTr), injecting local inductive bias via reconstruction. However, these methods lack explicit same-class contrastive constraints, making them sensitive to intraclass noise. Cross-domain frameworks like MCMD-CFSC [28] align image and text prototypes to enhance generalization, but dynamic interaction between spatial and spectral features remains underexplored.

Addressing these gaps, our spatial–spectral enhancement contrastive learning network (SSECLN) introduces MSE, bidirectional spectral modeling, and SCL, together with a dynamic cross-attention module and efficient PMC, enabling robust few-shot classification.

B. Contrastive Learning for Few-Shot HSI Classification

Contrastive learning has proven to be a powerful framework for few-shot learning by learning a representation space where samples of the same class are clustered together while being separated from dissimilar samples. This paradigm is particularly suitable for HSI classification with limited labels, as it can leverage the vast amount of unlabeled pixels and augmentations to learn generalizable features [29].

Contrastive learning is often integrated into the training process to regularize the feature extractor. A common strategy involves constructing positive and negative pairs through data augmentations (e.g., random cropping, band masking) and applying CL to the features extracted by an encoder network [30]. This approach encourages the model to learn augmentation-invariant and semantically meaningful representations. For instance, some methods employ a dual-branch Siamese network where different augmented views of the same pixel patch are pulled together in the feature space [15]. Others integrate contrastive objectives with masked image modeling, using a Transformer to reconstruct masked spectral–spatial tokens while a CL ensures consistency between global and local features [27]. A key component in these methods is the design of data augmentation strategies tailored for HSI. Effective augmentations must preserve the intrinsic spatial–spectral structure while creating meaningful variance. Strategies explored include spatial transformations (random cropping, flipping), spectral perturbations (band erasure, shuffling), and hybrid approaches [14], [31]. The quality of these augmentations directly impacts the effectiveness of the learned representations.

However, a limitation observed in several existing contrastive learning methods for HSI is their treatment of intraclass relationships. While they effectively separate different classes, they may not explicitly enforce sufficient compactness within a class, leaving the model sensitive to intraclass variations and noise, which is detrimental in the few-shot setting. Our proposed SSCL module directly addresses this by introducing an explicit SCL, which works in tandem with a novel spatial–spectral enhancement strategy to learn more discriminative and robust features from very few labeled examples.

TABLE I
STRUCTURE OF ENCODER AND DECODER

Encoder		Decoder	
SSCA	Conv^{QKV} + Attention	PosEmbed	Sin-Cos Embedding
1DConv	$3 \times 3 \times 1$	Block	MSA + MLP
Activation	Sigmoid	LayerNorm	LayerNorm
MeanPool	Global Mean Pool	Linear	decoder_embed_dim
PMC	Partial 1×1 Conv		
PatchEmbed	embed_dim		
PosEmbed	Sin-Cos Embedding		
Block	MSA + MLP		
LayerNorm	LayerNorm		

III. Proposed Methodology

The architecture of SSECLN is shown in Fig. 1. First, the SSCL is proposed, integrating MSE with SSE and BCL. The SSE module employs a dual-branch cooperative architecture: the MSE branch captures cross-scale spatial features through hierarchical convolutions, while the bidirectional spectral enhancement branch optimizes spectral representation via forward–backward propagation, ultimately generating discriminative spatial–spectral joint features. In parallel, the system concurrently incorporates two complementary modules: the SSCA establishes dynamic cross-dimensional interactions through dual-path attention mechanisms to excavate latent spatial–spectral correlations, while the PMC adopts dynamic convolution kernels and convolution positions to selectively focus on important local regions to reduce redundant spatial–spectral information. Finally, the BCL enhances feature discriminability by contrasting encoder–decoder outputs within each batch, optimized through a composite loss function that combines CL for encoder–decoder discrepancy minimization and SCL for intraclass feature alignment, enabling end-to-end network optimization.

A. Model Architecture

As shown in Fig. 1, the SSECLN is composed of an encoder and a decoder. We evenly divide the image into 27×27 patches, enhance features via the SSE module, refine the feature encoding–decoding structure, perform multiscale extraction and reconstruction, and introduce BCL reverse constraint supervision for parameter optimization.

As shown in Table I, the encoder adopts a dual-branch parallel structure to process input features. The first branch employs the SSCA module, which combines 1-D convolution and sigmoid to generate attention weights and performs global feature compression via global average pooling. The process is as follows:

$$\mathbf{F}_{\text{SSCA}} = \text{MeanPool}\left(\sigma\left(\text{1DConv}\left(\text{SSCA}\left(\mathbf{X}\right)\right)\right) \odot \mathbf{X}\right) \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ is the data enhanced by SSE, $\text{SSCA}(\cdot)$ is SSCA module, $\text{1DConv}(\cdot)$ is a $3 \times 3 \times 1$ convolution function, σ is the sigmoid function, $\odot$ is element-wise

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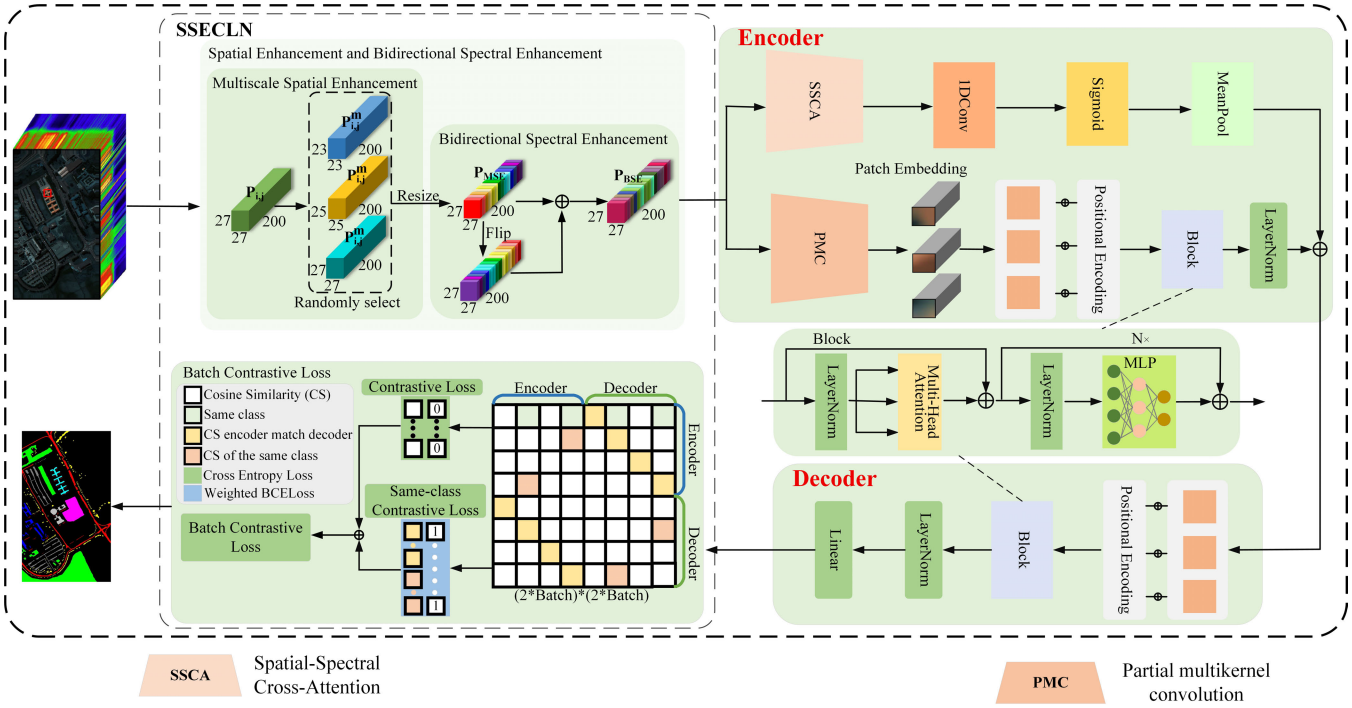


Fig. 1. Overview of the proposed SSECLN: 1) 27 × 27 patches are extracted from input HSIs; 2) MSE processes patches through parallel 23 × 23, 25 × 25 and 27 × 27 convolutional paths; 3) bidirectional spectral enhancement models both forward and reversed spectral contexts; and 4) features are optimized via batch CL, combining same CL and CL objectives to enhance discriminative power.

multiplication, and MeanPool$(\cdot)$ is global average pooling. In $\mathbb{R}^{H\times W\times C}$, C corresponds to the number of spectral channels, and H and W indicate the height and width of the input patch, respectively.

The second branch extracts local features via the PMC module, projects them via patch embedding, and incorporates positional encoding to provide positional information. The features are then iteratively refined by multiple Transformer blocks, where each block first normalizes the features, applies multihead attention, processes them through a residual MLP, and finally normalizes the output. Subsequently, the output features are fused with those from the SSCA branch to obtain complementary dual-path information. The process is as follows:

$$\begin{aligned}\mathbf{z}_0 &= \text{PatchEmbed}\left(\text{PMC}\left(\mathbf{X}\right)\right) + \mathbf{E}_{\text{pos}}\\ \mathbf{z}'_l &= \text{MSA}\left(\text{LayerNorm}\left(\mathbf{z}_{l-1}\right)\right) + \mathbf{z}_{l-1}\\ \mathbf{z}_l &= \text{MLP}\left(\text{LayerNorm}\left(\mathbf{z}'_l\right)\right) + \mathbf{z}'_l\\ \mathbf{z}_l &= \text{LayerNorm}\left(\mathbf{z}_l\right)\end{aligned} \tag{2}$$

where PMC$(\cdot)$ is PMC module, PatchEmbed$(\cdot)$ is the patch embedding function, $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{N\times D}$ is the learnable positional embedding, $N = (H \cdot W)/P^2$ is the number of patches, P is the patch size, D is the embedding dimension, LayerNorm$(\cdot)$ is layer normalization, MSA$(\cdot)$ denotes multihead attention, MLP$(\cdot)$ is the multilayer perceptron, l indexes the Transformer blocks, and z_{l-1}, z'_l, and z_l represent the input, intermediate, and output features of the lth block, respectively.

The decoder takes the encoder output tokens, adds positional encoding, and processes them through stacked Transformer blocks. The result is then refined by layer normalization and a linear projection layer to produce the final prediction map.

B. Spatial–Spectral Contrastive Learning

To better learn the diversity of spatial structures and continuity of spectral structures of ground objects in HSIs, MSE and bidirectional spectral enhancement are proposed. In addition, in few-shot data scenarios, due to the limited amount of training data, models fail to fully learn all data features, resulting in weak generalization ability. To address this issue, a batch contrastive learning method is proposed.

1) Spatial–Spectral Enhancement: The spatial structure of ground objects in HSIs has diversity, and using a single scale to extract features can lead to feature loss [32], [33], [34]. Therefore, we propose MSE as shown in Fig. 1, to extract multiscale spatial features and enhance the discriminability of spatial features. We define (i, j) as the index of pixels classified by the model. Then, extract a 27 × 27 patch centered around the classified pixels (i, j). This process can be expressed as follows:

$$\mathbf{P}_{i,j} = \mathbf{I}\left[i-13 : i+14,\ j-13 : j+14, :\right] \tag{3}$$

where $\mathbf{I}$ and $\mathbf{P}_{i,j}$, respectively, represent the images and patches extracted from the original HSI centered around (i, j).

Next, we randomly select a value from {23, 25, 27} to adjust its size, as follows:

$$\mathbf{P}_{\text{MSE}} = \text{resize}\left(\mathbf{P}^m_{i,j}, (27, 27)\right) \tag{4}$$

where m represents the randomly selected value of {23, 25, 27}; resize$(\mathbf{P}^m_{i,j}, (27, 27))$ represents the resizing of $\mathbf{P}^m_{i,j}$ to (27, 27),

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and $\mathbf{P}_{\mathrm{MSE}}$ represents the spatial characteristics after image enhancem ent by resizing operation.

For spectral characteristics, different objects may have the same spectrum, and the same object may exhibit different spectra under different backgrounds [35], [36]. However, existing unidirectional spectral modeling may miss highly discriminative local features due to local noise or nonlinear relationships between bands. To this end, we propose bidirectional spectral enhancement (SSE), as shown in Fig. 1, to capture the nonlinear relationships between spectral bands in both directions and avoid bias in unidirectional spectral modeling. We add $\mathbf{P}_{i,j}$ and flip($\mathbf{P}_{i,j}$) to obtain the enhanced spectral feature $\mathbf{P}_{\mathrm{BSE}}$ of the image

$$\mathbf{P}_{\mathrm{BSE}} = \mathbf{P}_{\mathrm{MSE}} + \mathrm{flip}\left(\mathbf{P}_{\mathrm{MSE}}\right) \tag{5}$$

where flip$(\cdot)$ represents the function that flips it along the spectral dimension.

2) Batch Contrastive Learning: In addition to using contrastive learning to bring the encoder and decoder results closer together [37], we introduce same-class sample constraints to alleviate their overfitting to local noise from similar training samples. Specifically, as shown in Fig. 1, first, the outputs of the encoder and decoder in the same batch are combined into a cosine similarity matrix of size (2 * Batch) × (2 * Batch). Then, the values corresponding to the encoder and decoder are combined with the values corresponding to the encoder and decoder of the same category. Finally, the loss function L_{scl} is calculated using a list of all 1 s in the same dimension. This process can be expressed as follows:

$$L_{scl} = -\left[y \cdot \log\left(\sigma\left(\frac{x}{\epsilon}\right)\right) + (1-y) \cdot \log\left(1-\sigma\left(\frac{x}{\epsilon}\right)\right)\right] \tag{6}$$

where $\mathbf{x} \in \mathbb{R}^{2\times \mathrm{Batch}+CS_{\mathrm{sc}}}$ represents the combination of the values corresponding to all encoders and decoders with the values corresponding to all encoders and decoders of the same class, $\mathbf{y} \in \mathbb{R}^{2\times \mathrm{Batch}+CS_{\mathrm{sc}}}$ represents the list of all ones of the same dimension as x, $\sigma(\cdot)$ represents the Sigmoid function, ϵ is a temperature hyperparameter, and CS_{sc} represents the number of cosine similarities of the same class in Fig. 1.

Next, we combine the residual values outside the decoder corresponding to the encoder of the cosine similarity matrix into x', and then use a loss function to calculate L_{cl} by combining x' and y'. This process can be expressed as follows:

$$L_{cl} = \sum_{i=1}^{(2*Batch)} \left[-\log\left(\frac{\exp\left(x'_{i,y'_i}\right)}{\sum_{j=1}^{(2*Batch-1)}\exp\left(x'_{i,j}\right)}\right)\right] \tag{7}$$

where $\mathbf{x}' \in \mathbb{R}^{(2*Batch)\times(2*Batch-1)}$ represents the combination of values corresponding to nonencoders and decoders and values of nonencoders and decoders of the same category. x' is a list of all zeros in the same dimension as $\mathbf{y}' \in \mathbb{R}^{(2*Batch)\times(2*Batch-1)}$, in order to keep nonencoders and decoders away from each other. $\sum_{j=1}^{(2*Batch-1)}\exp(x'_{i,j})$ is the sum of indices for all categories.

Finally, we combine L_{scl} and L_{cl} to obtain the final L_{bcl}. This process can be expressed as follows:

$$L_{bcl} = L_{scl} + L_{cl} \tag{8}$$

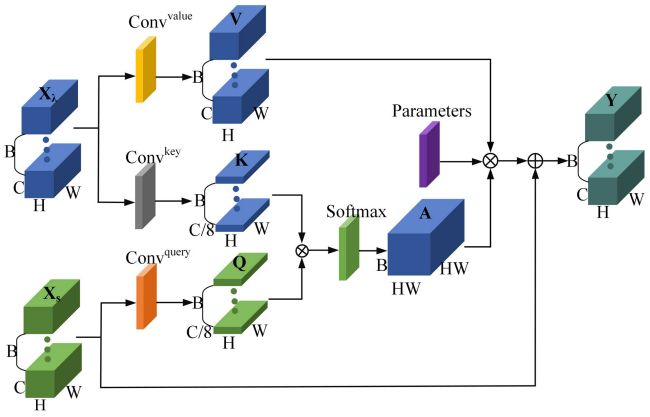


Fig. 2. Detailed architecture of the proposed SSCA.

where L_{bcl}, L_{scl}, and L_{cl} represent the batch-based CL, SCL, and CL, respectively.

C. Spatial–Spectral Cross-Attention

In this study, a novel SSCA is proposed that achieves adaptive fusion of local spatial structural information and global spectral dependency information by establishing a dynamic interaction mechanism between spatial and spectral dimensions. As shown in Fig. 2, the core of the SSCA is to construct a bidirectional attention interaction framework: assuming the input spatial feature is $\mathbf{X}_s \in \mathbb{R}^{B\times C\times H\times W}$ and the spectral feature is $\mathbf{X}_\lambda \in \mathbb{R}^{B\times C\times H\times W}$, in the spatial branch, fine-grained local spatial detail information Q is extracted from the high-resolution feature map through partial convolution; in the spectral branch, 1 × 1 partial convolutions $\mathrm{Conv}^{\mathrm{key}}$ and $\mathrm{Conv}^{\mathrm{value}}$ extract spectral features $\mathbf{K}$ and $\mathbf{V}$ from $\mathbf{X}_\lambda$, which encode global spectral dependencies across all bands. The process is expressed as

$$\begin{aligned}
\mathbf{Q} &= \mathrm{Conv}^{\mathrm{query}}\left(\mathbf{X}_s\right), \quad \mathbf{Q} \in \mathbb{R}^{B\times(C/8)\times H\times W} \\
\mathbf{K} &= \mathrm{Conv}^{\mathrm{key}}\left(\mathbf{X}_\lambda\right), \quad \mathbf{K} \in \mathbb{R}^{B\times(C/8)\times H\times W} \\
\mathbf{V} &= \mathrm{Conv}^{\mathrm{value}}\left(\mathbf{X}_\lambda\right), \quad \mathbf{V} \in \mathbb{R}^{B\times C\times H\times W}.
\end{aligned} \tag{9}$$

Here, $\mathrm{Conv}^{\mathrm{query}}, \mathrm{Conv}^{\mathrm{key}}, \mathrm{Conv}^{\mathrm{value}}$ denotes a partial convolution kernel with a kernel size of 1. $\mathbf{X}_s$ and $\mathbf{X}_\lambda$ are the identical spatial–spectral fused feature from SSE, serving as the spatial and spectral inputs to SSCA. In $\mathbb{R}^{B\times C\times H\times W}$, B represents the batch size, C corresponds to the number of spectral channels, and H and W indicate the height and width of the input patch, respectively.

Subsequently, the query feature $\mathbf{Q}$ and the key feature $\mathbf{K}$ are reshaped and matrix-multiplied to implement cross-attention calculation, and the obtained attention weight matrix $\mathbf{A}$ is normalized by Softmax to establish a dynamic long-range dependency relationship between local spatial positions and global spectral channels. This process can be expressed as follows:

$$\mathbf{A} = \mathrm{Softmax}\left(\mathrm{reshape}\left(\mathbf{Q}\right)\mathrm{reshape}\left(\mathbf{K}\right)\right), \quad \mathbf{A} \in \mathbb{R}^{B\times HW\times HW} \tag{10}$$

where reshape$(\cdot)$ represents the dimension reshaping function, which enables the model to automatically focus on spatially

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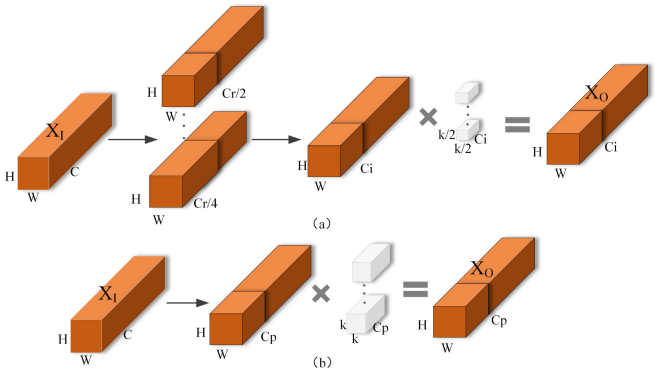


Fig. 3. (a) PMC. (b) Partial convolution.

discriminative regions and spectral information-rich channels, significantly improving the feature representation ability. Then, merge **A** and **V** through matrix multiplication, combine their outputs with learnable parameters, and finally add $\mathbf{X}_s$. This process can be expressed as follows:

$$\mathbf{Y} = \gamma \cdot \text{reshape}(\mathbf{V}) \cdot \text{reshape}(\mathbf{A}) + \mathbf{X}_s, \quad \gamma \in \mathbb{R} \tag{11}$$

where γ is a learnable parameter. Specifically, this module adopts a feature composition strategy to fuse the original input features with attention-weighted features, which preserves the low-level detail information and enhances the discriminative power of high-level semantic features. Experiments have shown that the SSCA can effectively model complex spatial–spectral coupling relationships by parallel processing of spatial–spectral features. The generated fusion features contain accurate descriptions of both local spatial structures and statistical properties of global spectral distributions. Compared with traditional serial feature fusion methods, this cross-attention mechanism can more fully explore the implicit correlations in spatial–spectral dimensions. In addition, the learnable parameters in the module are learned end-to-end, and the contribution weights of different modal features can be adaptively adjusted without manual intervention, significantly improving the flexibility of feature fusion while maintaining the lightweight of the model.

D. Partial Multikernel Convolution

CNN is widely used in HSIC tasks due to its ability to extract local features. As shown in Fig. 3(b), due to the need for multiple convolution operations in CNN, which frequently involve memory access, many researchers currently use partial convolution (PConv) in their own networks [26]. However, some convolution kernels and convolution positions are fixed, making it difficult to flexibly adapt to the feature extraction requirements of complex spatial–spectral information in remote sensing images. Therefore, in our network, we use PMC to more effectively extract the spectral and spatial information of HSI.

As shown in Fig. 3(a), PMC is a lightweight and efficient operation that can minimize frequent memory access and more effectively reduce floating-point operations (FLOPs). It first selects the position segments for partial convolution, and then selects multiskernel sizes based on the position segments, namely, the convolution kernel size, which can reduce redundant calculations of spatial–spectral information, as well as reduce memory access. Assuming we randomly select a convolution segment of length C_i from Cr/4 ••• Cr/2, and then use C_i to select the corresponding convolution kernel size $k' \times k'$ ($k' \in [k/4, 3k/4]$). The process is expressed as

$$k' = \left\lfloor \frac{k}{4}\left(\frac{8C_i}{C} - 1\right)\right\rfloor, \quad k' \in \left[\frac{k}{4}, \frac{3k}{4}\right] \tag{12}$$

where C represents the number of spectral channels, k represents the original convolution kernel size, C_i represents the length of a convolution segment, and $\lfloor\rfloor$ represents rounding down.

The input of a normal convolution is $\mathbf{X}_I \in \mathbb{R}^{H\times W\times C}$, and its generated output is $\mathbf{X}_O \in \mathbb{R}^{H\times W\times C}$, then the FLOPs and memory access count of this operation can be calculated as $H \times W \times k^2 \times C^2$ and $H \times W \times 2C + k^2 \times C^2$, while those for partial convolution are $H \times W \times k^2 \times C_p^2$ and $H \times W \times 2C_p + k^2 \times C_p^2$. In contrast, compared with normal convolution and partial convolution, PMC has lower FLOPs, calculated as $H \times W \times (k/2)^2 \times C_i^2\,(i \le p)$, and requires $H \times W \times 2C_i + (k/2)^2 \times C_i^2\,(i \le p)$ access count, where c_p and c_i represent the number of channels to be processed, which indicates that PMC has lower FLOPs and memory access count.

IV. Experimental Results and Discussion

In this section, we first describe the experimental setup, including the dataset, evaluation metrics, comparison methods, and implementation details. Then, from both qualitative and quantitative perspectives, we conducted a comprehensive comparison with state-of-the-art methods. Finally, we conducted a detailed ablation study to analyze the effectiveness of the proposed module.

A. Experimental Setting

Dataset: In order to more fairly demonstrate the effectiveness of SSAECL, we have selected four widely used and diverse hyperspectral datasets, namely, Indian Pines (IP), KSC, Salinas, and University of Pavia (UP).

1) *Indian Pines:* This dataset was collected in June 1992 by AVIRIS sensors at an IP agricultural testing site in northwest Indiana and contains 145 × 145 pixels with a spatial resolution of 20 × 20. Including 200 frequency bands and 16 land cover categories, with a total sample size of 10 249.
2) *KSC:* This dataset was collected by NASA's AVIRIS instrument in 1996. Its size is 512 × 614 × 176, with a spatial resolution of 18 m. After removing water absorption and low-SNR bands, 176 bands were used for analysis, resulting in a total sample size of 5211.
3) *Salinas:* This dataset was acquired by AVIRIS sensors in Salinas Valley, USA. This dataset consists of 512 × 217 pixels with a spatial resolution of 3.7 m/pixel. After removing 20 water-absorbing bands, there are a total of 204 frequency bands with a total sample size of 54 129, involving 16 land cover categories.

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4) *University of Pavia:* This dataset was captured by the Reflective Optical System Imaging Spectrometer (ROSIS) located in Pavia, northern Italy. The size of the data is 610 × 340, with 103 spectral bands and a spectral coverage range of 0.43–0.86 μm. The UP dataset contains 42 776 labeled pixels for nine city categories.
   a) *Evaluation metric:* All indicators of this experiment were evaluated on the test set. We use three widely used evaluation metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). To eliminate the bias caused by randomly selecting training samples, the experiment was repeated 10 times, and the average and standard deviation of the evaluation indicators were statistically analyzed. Therefore, the higher the average value of these evaluation indicators and the smaller the standard deviation, the better the performance of the model.
   b) *Comparison methods:* We compared relevant HSI classification methods.
      1) *SSSAN [TGRS22] [38]:* This model consists of two subnetworks, spatial and spectral, each embedded with self-attention modules to capture correlations, and also uses "score-weighted" fusion to extract spatial–spectral features.
      2) *SSTN [TGRS22] [39]:* This model includes spatial attention and spectral correlation modules, which use the FAS framework to determine hierarchical operations and block-level order, resulting in low computational costs.
      3) *SSFTT [TGRS22] [40]:* This model includes spectral–spatial feature extraction, a Gaussian-weighted feature tokenizer, and a Transformer encoder, and uses a linear layer to identify labels.
      4) *CVSSN [TCSVT22] [41]:* This model is based on a self-similarity network oriented toward center vectors, and uses similarity metrics to mine spatial associations between centers and neighbors in order to optimize spectral–spatial feature extraction.
      5) *AMS-M2ESL [TGRS23] [42]:* This model enhances the spectral–spatial feature extraction capability of HSIs through adaptive mask sampling, distance covariance representation, and manifold-to-Euclidean space projection techniques.
      6) *SC-SS-MTr [TGRS23] [27]:* This model includes a basic spectral–spatial mask Transformer and three derivative models. Through two-stage training, local inductive bias is injected from mask input reconstruction, and additional branches are used to fuse contrastive learning and supervised learning.
      7) *SQSFormer [TGRS24] [25]:* This model is based on a Transformer and improves the classification performance of few-shot HSIs by adaptively fusing spectral–spatial information through rotation-invariant position embedding and spectral–spatial center attention.
      8) *SSEFN [TGRS25] [24]:* This model improves feature discriminative power through the spatial–spectral enhancement strategy, combined with an adaptive decision fusion module to alleviate overfitting and enhance the classification performance of few-shot HSIs.
      9) *MCTGCL [TGRS25] [26]:* This model combines CNN and Transformer, extracts features through an information-enhanced attention module and a lightweight dual-branch network, introduces graph contrastive learning, and improves the classification performance of few-shot Mars HSIs.
      10) *AdaMixCon [TGRS26] [43]:* This Mamba-based two-stage framework for few-shot HSI classification employs HSI-MixCon for spectral–spatial feature learning and DASR for difficulty-aware sample routing to boost few-shot HSI classification performance.

   c) *Implementation details:* The proposed SSECLN is implemented on the PyTorch platform, with all runs running on the same hardware: NVIDIA GEFORCE RTX 3090 GPU, 64 Intel Core i5-13400 CPU, and 32 GB RAM. All experiments were repeated ten times, randomly selecting five samples from each labeled dataset as the training and validation sets, and then selecting the remaining test sets. Detailed quantities are provided in Table II. We use the AdamW optimizer, as shown in Fig. 4, with training episodes of 180, a learning rate initialized to 0.0001, and a temperature hyperparameter of 1. As shown in Fig. 5, we conducted experiments on the IP, KSC, Salinas, and UP datasets by fixing all other hyperparameters while varying the batch size, patch size, kernel size, and C size, respectively. For the batch size, we tested the values of 6, 8, 10, 12, 14, and 16; the results demonstrate that a batch size of ten yields the highest overall accuracy (OA) across all four datasets. For the patch size, we evaluated the configurations of 9×9, 15×15, 21×21, 27×27, and 33×33, and the experimental outcomes reveal that a patch size of 27×27 achieves the optimal classification performance on all tested datasets. Regarding the kernel size, we examined five different configurations ([0, k], [k/2, k/2], [k/3, 2k/3], [k/4, 3k/4], [k/5, 4k/5]), and the results indicate that the [k/4, 3k/4] setting consistently delivers the best OA across all datasets. Regarding the C size, we examined five different configurations ([0, C_r/4], [C_r/4, C_r/2], [0, C_r/2], [C_r/2, C_r], [3C_r/4, C_r]), and the results indicate that the [C_r/4, C_r/2] setting consistently delivers the best OA across all datasets.

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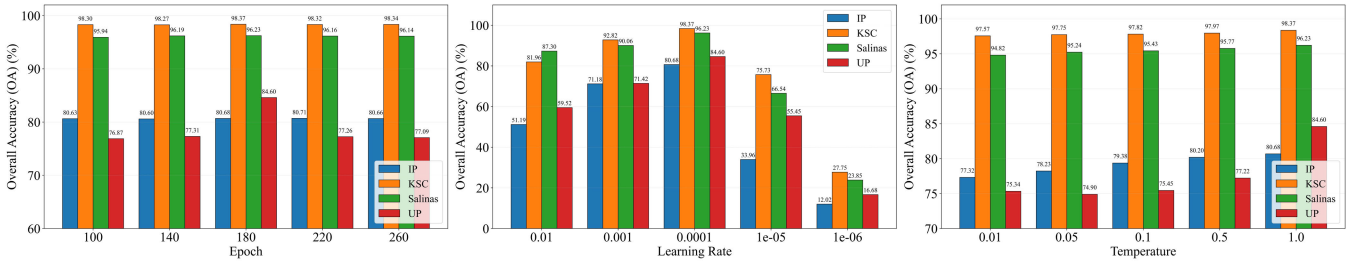


Fig. 4. Parameter tuning of epoch, learning rate, and ϵ for the proposed network in the four datasets.

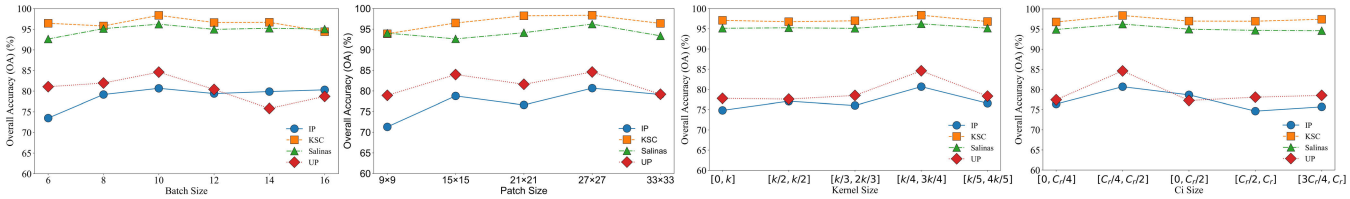


Fig. 5. Effect of batch, patch, kernel, and C_i sizes on the OA metrics in the four datasets.

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR IP, KSC, SALINAS, AND UP DATASETS

No.	Indian Pines (IP)				KSC				Salinas				University of Pavia (UP)			
	Class	Train	Val	Test	Class	Train	Val	Test	Class	Train	Val	Test	Class	Train	Val	Test
1	Alfalfa	5	5	36	Scrub	5	5	751	Brocoli_green_weeds_1	5	5	1999	Asphalt	5	5	6621
2	Corn-notill	5	5	1418	Willow swamp	5	5	233	Brocoli_green_weeds_2	5	5	3716	Meadows	5	5	18639
3	Corn-mintill	5	5	820	Cabbage palm	5	5	246	Fallow	5	5	1966	Gravel	5	5	2099
4	Corn	5	5	227	Cabbage palm/oak	5	5	242	Fallow_rough_plow	5	5	1384	Trees	5	5	3054
5	Grass-pasture	5	5	473	Slash pine	5	5	151	Fallow_smooth	5	5	2668	Metal Sheets	5	5	1335
6	Grass-trees	5	5	720	Oak/broadleaf	5	5	219	Stubble	5	5	3949	Bare soil	5	5	5019
7	Grass-pasture-mowed	5	5	18	Hardwood swamp	5	5	95	Celery	5	5	3569	Bitumen	5	5	1320
8	Hay-windrowed	5	5	468	Graminoid marsh	5	5	421	Grapes_untrained	5	5	11261	Bricks	5	5	3672
9	Oats	5	5	10	Spartina marsh	5	5	510	Soil_vinyard_develop	5	5	6193	Shadows	5	5	937
10	Soybean-notill	5	5	962	Cattail marsh	5	5	394	Corn_senesced_green_weeds	5	5	3268				
11	Soybean-mintill	5	5	2445	Salt marsh	5	5	409	Lettuce_romaine_4wk	5	5	1058				
12	Soybean-clean	5	5	583	Mud flats	5	5	493	Lettuce_romaine_5wk	5	5	1917				
13	Wheat	5	5	195	Water	5	5	917	Lettuce_romaine_6wk	5	5	906				
14	Woods	5	5	1255					Lettuce_romaine_7wk	5	5	1797				
15	Buildings	5	5	376					Vinyard_vertical_trellis	5	5	1797				
16	Stone-Steel-Towers	5	5	83												
Total		**80**	**80**	**10089**		**65**	**65**	**5081**		**80**	**80**	**54129**		**45**	**45**	**42786**

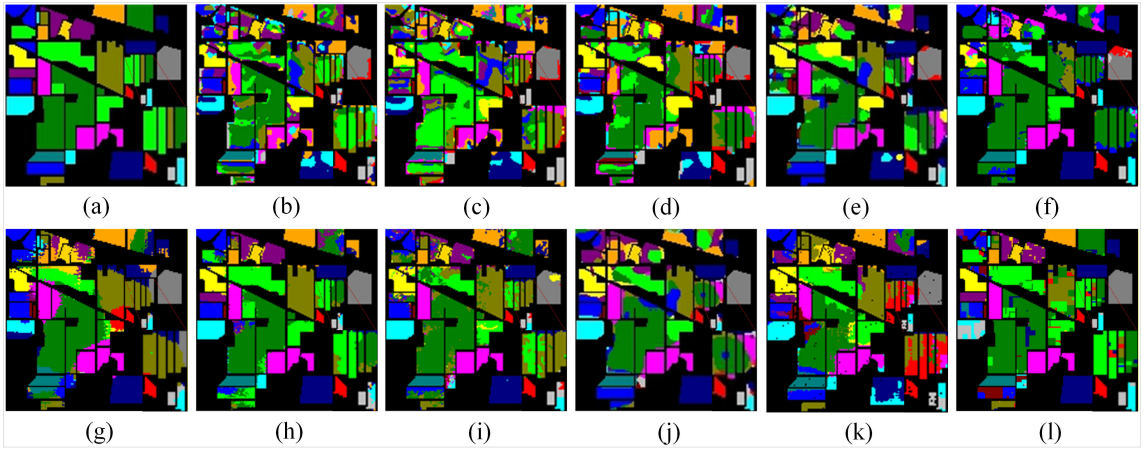


Fig. 6. Qualitative visualization of the classification map for the IP dataset. (a) Ground-truth map. (b) SSSAN. (c) SSTN. (d) SSFTT. (e) CVSSN. (f) AMS-M2ESL. (g) SC-SS-MTr. (h) SQSFormer. (i) SSEFN. (j) MCTGCL. (k) AdaMixCon. (l) Proposed SSECLN.

B. Comparison With State-of-the-Arts

In order to fairly evaluate the performance and efficiency of the proposed model, we comprehensively compared our method with state-of-the-art HSI classification methods from both quantitative and qualitative perspectives. Quantitative and qualitative evaluations are reported in Tables III–VI

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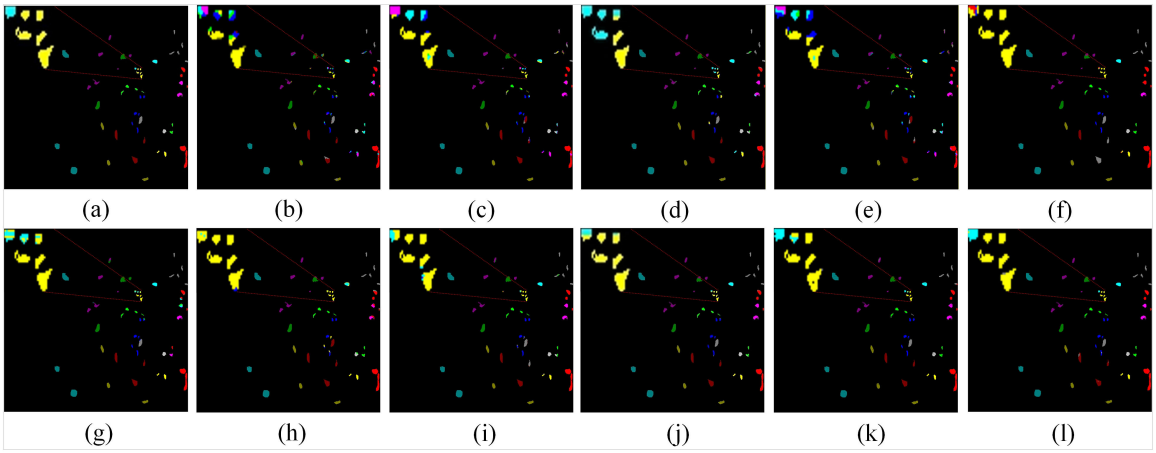


Fig. 7. Qualitative visualization of the classification map for the KSC dataset. (a) Ground-truth map. (b) SSSAN. (c) SSTN. (d) SSFTT. (e) CVSSN. (f) AMS-M2ESL. (g) SC-SS-MTr. (h) SQSFormer. (i) SSEFN. (j) MCTGCL. (k) AdaMixCon. (l) Proposed SSECLN.

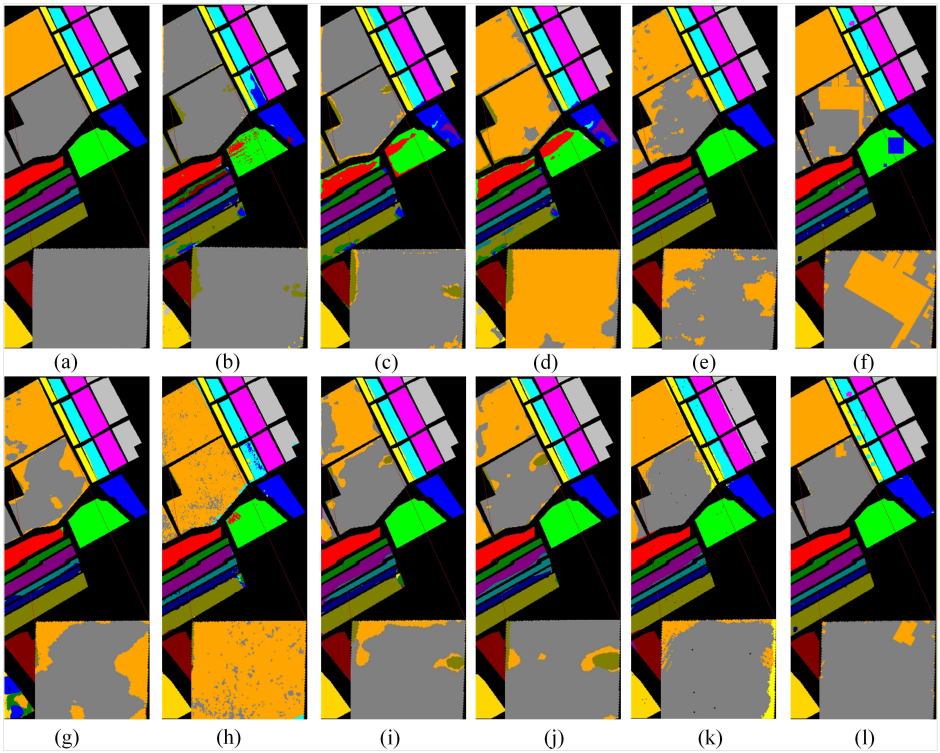


Fig. 8. Qualitative visualization of the classification map for the Salinas dataset. (a) Ground-truth map. (b) SSSAN. (c) SSTN. (d) SSFTT. (e) CVSSN. (f) AMS-M2ESL. (g) SC-SS-MTr. (h) SQSFormer. (i) SSEFN. (j) MCTGCL. (k) AdaMixCon. (l) Proposed SSECLN.

and Figs. 6–9. From the quantitative results, we found that SQSFormer, SSEN, MCTGCL, and AdaMixCon are few-shot methods with higher accuracy than other methods. Taking the KSC dataset as an example, AdaMixCon performed better than the most competitive SC-SS-MTr with 5.65%, 7.23%, and 6.28%, respectively, in OA, AA, and Kappa. This may be due to overfitting caused by other methods not considering the situation where there are almost no training samples. More importantly, compared to other models in all four datasets, our proposed SSECLN model performs the best. This proves that our method has good adaptability to various environments. Taking the IP dataset as an example, the proposed SSECLN model performs better than the most competitive AdaMixCon, accounting for 6.12%, 1.96%, and 6.76% in OA, AA, and Kappa, respectively. This is because AdaMixCon is limited to a single scale for feature extraction, while SSECLN not only captures multiscale spatial information but also considers bidirectional spectral information, resulting in better performance. Due to the uneven sample size in the IP dataset, many categories with few-shot sizes experience performance fluctuations. In addition, the same type constraints of BCL are difficult to effectively implement, and the feature extraction consistency of PMC is low, resulting in very high standard deviations in some basic categories, such as category 7 and

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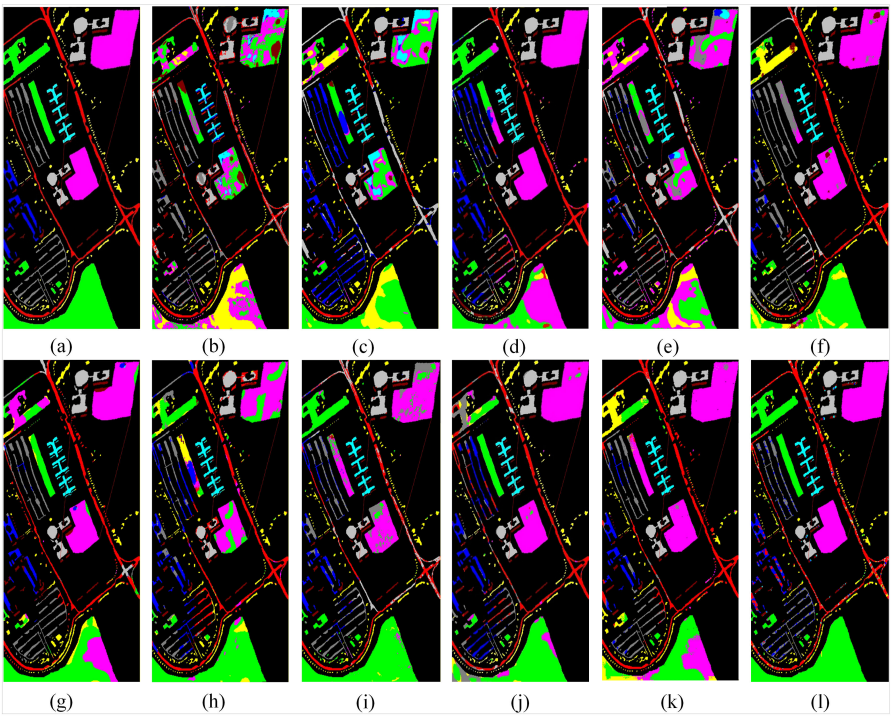


Fig. 9. Qualitative visualization of the classification map for the UP dataset. (a) Ground-truth map. (b) SSSAN. (c) SSTN. (d) SSFTT. (e) CVSSN. (f) AMS-M2ESL. (g) SC-SS-MTr. (h) SQSFormer. (i) SSEFN. (j) MCTGCL. (k) AdaMixCon. (l) Proposed SSECLN.

TABLE III

QUANTITATIVE RESULT (ACC% ± STD) OF THE IP DATASET. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	SSSAN TGRS2022	SSTN TGRS2022	SSFTT TGRS2022	CVSSN TCSVT2022	AMS-M2ESL TGRS2023	SC-SS-MTr TGRS2023	SQSFormer TGRS2024	SSEFN TGRS2025	MCTGCL TGRS2025	AdaMixCon TGRS2026	SSECLN Ours
Alfalfa	27.93±12.02	32.66±20.17	36.49±13.96	38.89±23.45	54.56±39.45	17.30±2.27	58.56±10.84	51.83±15.67	56.11±31.29	**100.00±0.00**	32.95±3.58
Corn-notill	46.61±19.90	48.91±14.17	62.26±3.81	45.98±16.38	77.71±28.19	**83.88±6.35**	71.76±15.48	69.22±13.85	73.21±13.98	63.42±5.67	79.33±3.42
Corn-mintill	26.19±10.11	37.81±13.20	57.61±8.56	32.05±10.55	67.56±32.42	66.99±4.41	**79.28±11.97**	54.12±15.41	69.19±9.01	55.92±11.53	74.72±4.18
Corn	26.39±10.16	42.61±10.26	38.09±8.18	46.78±16.75	55.87±28.05	60.86±10.87	58.99±18.37	44.86±15.70	64.01±14.24	**95.99±6.79**	81.14±5.63
Grass-pasture	44.90±9.05	52.25±13.18	77.31±5.38	45.52±10.06	95.64±7.44	81.93±6.54	84.21±12.20	86.76±9.47	83.34±6.73	75.43±8.54	**97.74±2.65**
Grass-trees	62.11±13.08	75.24±15.78	87.69±5.93	84.92±5.29	90.55±19.76	70.50±5.07	86.77±6.90	**97.28±1.66**	89.51±5.30	87.29±9.10	83.53±1.84
Grass-pasture-mowed	21.18±16.67	15.62±5.95	36.00±11.31	20.63±10.13	87.24±20.54	43.81±6.43	16.82±6.43	22.65±8.01	49.97±24.67	**100.00±0.00**	45.79±5.13
Hay-windrowed	91.98±6.97	98.39±2.12	98.06±2.14	94.49±0.51	97.05±8.63	81.53±2.79	98.03±2.90	96.72±1.91	97.99±2.80	**99.64±0.72**	97.62±2.85
Oats	9.91±8.16	2.82±0.77	14.54±6.62	6.15±4.68	64.97±35.57	13.92±6.28	20.92±7.48	23.73±7.03	34.73±13.11	**100.00±0.00**	8.62±8.21
Soybean-notill	38.71±7.16	48.05±17.24	71.91±9.32	53.57±14.91	58.21±28.86	67.50±7.32	71.88±13.06	56.52±12.74	65.31±6.86	74.32±0.06	**78.92±2.11**
Soybean-mintill	63.55±22.98	66.93±13.64	76.74±5.98	67.64±12.85	76.91±20.55	87.04±3.70	85.45±6.53	72.31±3.93	82.31±9.37	64.97±11.87	**89.92±1.73**
Soybean-clean	21.97±5.63	28.09±5.47	36.13±4.47	25.42±3.91	62.76±20.15	**84.11±8.70**	70.19±11.71	48.53±11.47	54.48±14.06	72.90±10.20	80.34±5.41
Wheat	65.38±14.92	69.92±17.01	83.36±3.91	75.99±12.40	96.87±6.81	52.54±6.91	55.61±13.01	89.16±6.15	69.51±3.66	**99.30±1.93**	78.64±3.82
Woods	81.20±7.94	85.72±2.09	84.09±5.49	89.57±4.97	**98.28±4.14**	80.76±6.87	94.93±6.50	96.87±1.92	96.41±3.35	85.46±8.51	92.81±1.30
Buildings-Grass-Trees-Drives	38.10±10.07	45.96±15.73	67.88±10.31	49.77±6.66	63.80±29.92	59.65±8.39	66.32±16.75	74.16±11.17	64.69±9.99	**91.67±9.05**	83.42±4.38
Stone-Steel-Towers	58.00±20.27	68.09±11.49	59.52±5.67	80.73±8.21	72.34±33.57	29.73±5.79	59.30±19.84	67.78±16.14	63.37±15.95	**100.00±0.00**	37.76±3.80
OA(%)	46.09±6.11	51.68±5.43	68.30±1.28	54.16±4.12	61.52±6.36	71.50±2.10	73.99±2.21	74.49±4.66	74.82±4.23	74.56±3.66	**80.68±0.27**
AA(%)	45.26±4.83	51.18±2.73	80.03±0.56	53.64±3.08	76.27±8.84	81.28±1.38	84.97±0.77	70.76±3.92	85.17±2.87	85.39±1.87	**87.35±0.28**
Kappa(%)	40.21±6.18	46.37±5.61	64.37±1.33	48.77±4.48	57.07±6.17	68.08±2.26	70.86±2.35	71.33±4.96	71.69±4.69	71.56±3.87	**78.32±0.28**

category 9. Furthermore, grass trees, soybean-mintill, and woods have sufficient sample sizes and continuous spatial distributions, with high discriminability in both spectral and spatial features, which typically results in lower standard deviations.

From the qualitative evaluation, the classification graphs generated by the corresponding methods in the IP, KSC, Salinas, and UP datasets are shown in Figs. 6–9. Overall, compared to other methods, the proposed SSECLN model has less misclassification, resulting in high-quality classification maps with smooth objects and well-maintained boundaries. Taking the UP dataset as an example, all comparison methods contain many noisy areas, misclassified areas, and blurred boundaries. On the contrary, the proposed SSECLN model exhibits better object continuity and clearer boundaries.

In summary, our proposed SSECLN achieves the best classification performance among all quantitative metrics and visualizations compared to state-of-the-art individuals. This demonstrates the advantages of the proposed SSECLN.

C. Ablation Study

In order to further explore and validate the contributions of different modules of the proposed SSECLN model, ablation experiments were conducted on four highly discriminative datasets, IP, KSC, Salinas, and UP, with the same parameter

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TABLE IV

QUANTITATIVE RESULT (ACC% ± STD) OF THE KSC DATASET. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	SSSAN TGRS2022	SSTN TGRS2022	SSFTT TGRS2022	CVSSN TCSVT2022	AMS-M2ESL TGRS2023	SC-SS-MTr TGRS2023	SQSFormer TGRS2024	SSEFN TGRS2025	MCTGCL TGRS2025	AdaMixCon TGRS2026	SSECLN Ours
Scrub	94.58±4.32	98.76±1.77	97.59±3.91	96.59±3.55	**99.92±0.17**	96.44±2.15	99.56±0.79	98.27±1.54	98.98±1.40	99.58±0.91	99.17±1.22
Willow swamp	59.21±17.31	74.47±18.59	94.81±4.55	65.24±15.61	95.75±9.73	78.50±14.09	94.76±7.32	96.28±2.80	95.87±4.51	94.07±10.88	**99.06±2.58**
Cabbage palm	61.29±15.76	87.62±6.69	73.16±18.16	83.25±17.06	77.13±21.48	93.36±4.86	86.10±10.02	90.61±4.79	92.56±6.33	98.56±4.17	**99.03±2.07**
Cabbage palm/oak	49.30±20.15	52.37±12.02	92.73±9.37	58.42±12.42	85.57±29.71	84.64±12.05	95.64±6.04	90.71±6.75	93.42±4.65	95.34±5.79	**99.05±1.30**
Slash pine	41.45±13.31	52.83±12.35	90.70±14.78	45.17±13.37	86.44±21.88	89.51±5.88	96.91±5.30	91.91±6.23	88.07±10.03	90.64±9.23	**97.69±2.17**
Oak/broadleaf	60.08±14.02	70.13±9.77	86.66±10.28	74.39±13.59	72.53±24.53	72.58±15.99	87.62±22.26	88.77±9.85	94.10±10.93	**96.33±10.98**	94.53±4.68
Hardwood swamp	69.60±20.58	74.20±7.43	76.39±11.59	81.30±18.32	67.15±17.53	87.88±12.81	84.92±15.18	79.18±11.26	84.95±14.53	**100.00±0.00**	96.62±7.04
Graminoid marsh	52.73±9.82	62.14±9.01	90.25±7.77	61.75±7.39	82.07±29.60	90.51±4.47	95.88±6.92	92.17±7.44	90.93±7.79	95.44±8.96	**97.79±2.48**
Spartina marsh	81.48±10.89	84.60±7.42	89.74±7.59	88.66±8.97	87.09±19.46	**100.00±0.00**	99.92±0.17	98.34±2.28	99.46±1.12	91.84±8.15	**100.00±0.00**
Cattail marsh	64.75±14.62	85.56±9.64	99.85±0.31	81.49±8.84	99.59±1.22	**100.00±0.00**	**100.00±0.00**	99.17±0.99	**100.00±0.00**	**100.00±0.00**	99.61±0.85
Salt marsh	94.77±8.38	95.73±7.03	**99.67±0.35**	95.07±8.99	94.77±12.29	81.55±5.22	95.36±4.60	99.43±0.55	99.30±1.06	97.41±3.89	92.25±4.65
Mud flats	88.68±11.44	88.30±8.15	94.25±3.04	95.42±9.99	93.89±3.98	**100.00±0.00**	95.42±4.05	94.62±4.12	95.68±3.17	98.87±2.42	**100.00±0.00**
Water	92.62±7.65	98.71±1.38	99.84±0.19	96.54±5.02	**100.00±0.00**	99.77±0.42	**100.00±0.00**	**100.00±0.00**	99.81±0.47	**100.00±0.00**	**100.00±0.00**
OA(%)	74.09±5.07	84.11±1.91	92.37±1.92	82.34±2.84	88.18±5.22	91.76±1.28	95.86±2.52	95.26±0.90	96.28±1.51	97.41±1.32	**98.37±0.26**
AA(%)	69.66±3.49	78.88±2.18	90.50±2.29	77.95±3.89	87.84±4.76	89.54±1.25	95.13±2.10	93.04±1.43	95.20±2.00	96.77±1.47	**98.56±0.27**
Kappa(%)	71.14±5.58	82.31±2.12	91.54±2.13	80.31±3.17	86.87±5.80	90.84±1.41	95.40±2.79	94.72±1.00	95.86±1.68	97.12±1.47	**98.21±0.29**

TABLE V

QUANTITATIVE RESULT (ACC% ± STD) OF THE SALINAS DATASET. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	SSSAN TGRS2022	SSTN TGRS2022	SSFTT TGRS2022	CVSSN TCSVT2022	AMS-M2ESL TGRS2023	SC-SS-MTr TGRS2023	SQSFormer TGRS2024	SSEFN TGRS2025	MCTGCL TGRS2025	AdaMixCon TGRS2026	SSECLN Ours
Brocoli_green_weeds_1	51.72±23.66	62.16±18.33	98.18±2.42	52.59±19.52	97.33±4.83	98.34±1.90	**100.00±0.00**	98.88±2.25	99.93±0.14	99.97±0.04	99.55±0.46
Brocoli_green_weeds_2	83.39±14.81	90.83±7.79	97.71±2.39	94.19±5.42	97.15±5.86	**100.00±0.00**	99.27±0.10	99.75±0.29	99.48±1.12	99.75±0.63	99.59±0.45
Fallow	76.21±14.54	84.12±7.74	94.26±11.90	86.09±10.00	**99.73±0.64**	98.83±0.60	86.47±13.36	96.94±0.81	99.39±1.12	**99.73±0.33**	99.54±0.53
Fallow_rough_plow	91.12±7.55	80.09±11.06	94.65±3.48	87.70±6.51	95.95±2.29	79.27±7.36	95.03±4.96	96.58±2.86	93.12±4.81	**98.61±2.68**	96.58±1.48
Fallow_smooth	93.31±3.06	96.23±3.49	97.72±1.77	94.01±7.12	94.69±8.27	93.33±2.22	**98.52±1.01**	94.60±3.79	95.98±4.83	94.10±3.20	97.80±1.23
Stubble	99.17±1.94	99.38±1.06	97.90±1.60	99.66±0.77	**99.91±0.24**	86.47±1.66	99.28±0.63	99.90±0.08	98.87±1.45	97.60±2.44	95.37±1.79
Celery	91.73±7.80	95.16±3.74	98.68±1.04	95.65±2.33	97.87±2.99	94.55±1.77	99.98±0.01	98.94±1.62	99.72±0.45	98.91±1.98	**100.00±0.00**
Grapes_untrained	70.15±13.73	70.88±8.29	82.68±6.16	76.95±13.22	87.11±14.20	84.93±2.63	91.66±1.93	83.16±2.75	90.38±6.93	83.81±9.35	**99.52±0.29**
Soil_vinyard_develop	92.81±3.00	95.46±2.79	93.63±12.68	98.00±2.57	94.60±8.20	99.97±0.03	99.51±0.32	98.95±0.61	98.55±1.20	99.89±0.29	**100.00±0.00**
Corn_senesced_green_weeds	81.44±8.71	81.41±10.32	89.06±7.15	82.24±7.88	95.40±7.86	94.28±2.56	97.97±0.63	96.46±1.59	94.39±4.48	93.80±7.01	**99.95±0.06**
Lettuce_romaine_4wk	40.01±13.68	53.13±8.14	88.87±5.44	59.74±10.66	97.65±3.73	90.14±7.12	81.90±14.28	92.75±1.34	93.75±4.62	**99.85±0.24**	98.68±0.74
Lettuce_romaine_5wk	80.40±12.29	87.08±7.75	90.77±7.47	87.21±11.18	98.34±1.76	99.41±1.41	99.60±0.23	98.89±0.60	96.31±4.12	95.73±5.33	**100.00±0.00**
Lettuce_romaine_6wk	66.42±9.44	87.40±16.42	82.95±11.37	74.50±14.31	80.39±30.44	**99.40±0.91**	95.77±1.78	95.92±8.27	84.51±15.92	97.61±4.52	98.89±0.50
Lettuce_romaine_7wk	82.34±13.60	81.59±12.52	87.83±6.39	87.61±10.16	81.66±28.35	66.94±2.67	79.77±0.85	94.10±2.34	93.80±4.66	96.97±4.52	**98.36±0.48**
Vinyard_untrained	50.47±10.43	47.78±9.56	67.81±10.26	51.99±7.02	55.55±2.97	79.61±1.37	68.64±2.83	72.25±6.05	77.39±6.19	**80.56±6.16**	70.85±1.03
Vinyard_vertical_trees	75.25±18.74	84.62±16.63	96.88±5.80	86.13±7.03	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.73±0.37	99.64±0.64	99.88±0.23	99.94±0.11
OA(%)	72.83±2.54	76.31±4.02	87.51±2.84	75.94±2.64	82.50±5.45	90.13±0.52	90.82±0.20	91.10±0.66	92.56±0.85	93.84±1.97	**96.23±0.06**
AA(%)	76.64±3.41	81.09±2.97	93.39±1.33	82.15±1.33	92.09±3.31	92.75±0.54	92.11±1.67	94.47±0.54	95.46±1.28	96.55±1.10	**97.82±0.09**
Kappa(%)	69.91±2.72	73.79±4.31	86.17±3.10	73.50±2.78	80.74±5.80	89.01±0.59	89.80±0.22	90.10±0.72	91.73±0.96	93.16±2.17	**95.91±0.07**

TABLE VI

QUANTITATIVE RESULT (ACC% ± STD) OF THE UP DATASET. BEST IN BOLD AND SECOND WITH UNDERLINE

Class	SSSAN TGRS2022	SSTN TGRS2022	SSFTT TGRS2022	CVSSN TCSVT2022	AMS-M2ESL TGRS2023	SC-SS-MTr TGRS2023	SQSFormer TGRS2024	SSEFN TGRS2025	MCTGCL TGRS2025	AdaMixCon TGRS2026	SSECLN Ours
Asphalt	78.95±12.49	81.45±11.66	86.01±3.09	86.40±6.59	85.59±13.33	85.37±3.03	73.61±5.95	**98.71±0.90**	90.39±6.39	78.56±6.28	92.25±1.89
Meadows	74.29±16.34	76.27±26.30	94.93±2.00	87.83±3.53	93.85±4.33	**98.17±0.78**	90.44±4.71	93.22±2.64	96.35±2.66	71.76±14.09	95.71±1.06
Gravel	29.48±12.34	25.09±10.64	42.17±4.36	36.51±16.47	52.43±21.08	**88.60±7.90**	68.03±19.75	63.86±13.61	60.38±13.58	79.53±6.78	60.56±3.32
Trees	47.84±24.55	28.15±10.70	78.49±6.48	58.43±17.02	46.73±17.74	41.69±7.98	77.74±18.69	80.81±18.78	66.90±16.76	73.15±6.84	**85.83±4.15**
Metal Sheets	77.38±23.95	82.07±18.01	90.54±3.26	92.97±5.03	99.90±0.04	85.96±0.45	89.22±2.30	**99.97±0.03**	95.77±5.19	99.95±0.08	86.02±2.36
Bare Soil	32.68±10.08	35.17±14.20	39.09±2.52	37.14±8.53	83.03±12.21	54.12±7.23	78.70±15.94	74.06±14.52	67.11±10.90	**91.08±10.91**	72.30±6.82
Bitumen	28.56±10.19	31.97±10.60	75.01±13.40	41.22±10.73	53.93±23.36	66.49±5.81	89.73±12.04	38.14±9.43	74.54±17.90	**99.54±0.60**	59.60±5.02
Bricks	63.58±10.49	42.40±16.64	84.77±5.90	52.31±19.11	41.57±13.37	**87.03±3.72**	61.57±6.34	80.70±6.55	55.48±14.31	83.18±9.76	86.76±1.64
Shadows	48.76±22.60	52.63±21.35	62.47±14.10	84.75±13.59	65.54±9.44	63.49±7.03	57.46±10.33	**90.64±7.69**	77.64±12.78	89.06±7.22	85.98±2.51
OA(%)	53.35±9.66	48.75±10.35	59.88±2.05	64.59±6.54	67.49±7.17	75.29±3.28	78.74±2.76	80.74±3.86	78.67±3.42	78.68±6.40	**84.60±0.47**
AA(%)	53.51±5.59	50.58±9.18	67.08±1.89	64.18±3.26	69.18±4.56	83.84±1.06	75.86±2.54	80.01±2.87	83.70±2.64	85.09±2.71	**86.63±0.64**
Kappa(%)	43.42±8.83	39.81±9.59	79.31±0.87	56.21±6.71	60.31±7.79	69.57±3.64	72.00±2.74	75.22±4.52	73.05±3.94	73.39±7.40	**81.66±0.55**

configuration to analyze the effectiveness of each component, including SSCL, SSCA, and PMC. As recorded in Table VII, we can observe that follows.

1) The proposed SSCL is much better than simply fixed spatial–spectral scales. It can better extract multi-scale spatial information and fully explore spectral dependencies. It also has a better performance than using SimCLR, indicating that it can better alleviate model overfitting and improve model generalization ability. The experimental results prove the effectiveness of introducing SSCL. It can be observed that the accuracy of the four datasets will be improved after introducing this

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TABLE VII

CLASSIFICATION PERFORMANCE (%) OF THE ABLATIONS ON THE PROPOSED METHOD ON IP, KSC, SALINAS, AND UP

Components			Metrics	Datasets			
SSCL	SSCA	PMC		IP	KSC	Salinas	UP
			OA	70.66±1.31	96.68±0.53	94.53±0.66	74.43±2.70
✗	✗	✗	AA	80.59±0.69	96.92±0.56	96.46±0.42	78.04±1.74
			Kappa	67.33±1.42	96.35±0.59	94.07±0.71	69.87±3.04
			OA	77.49±1.24	96.72±0.67	94.65±0.59	79.14±1.80
✓	✗	✗	AA	85.61±0.62	97.13±0.62	96.40±0.50	81.69±1.33
			Kappa	74.83±1.37	96.39±0.74	94.21±0.64	75.25±2.07
			OA	71.67±1.54	96.84±0.50	94.87±0.48	75.07±2.14
✗	✓	✗	AA	81.43±0.80	97.20±0.45	96.80±0.35	78.93±1.49
			Kappa	68.41±1.68	96.52±0.55	94.44±0.52	70.61±2.42
			OA	73.96±2.00	96.82±0.32	95.25±0.48	76.21±2.80
✗	✗	✓	AA	82.59±1.08	97.12±0.36	97.00±0.38	80.36±1.76
			Kappa	70.92±2.16	96.50±0.35	94.86±0.51	72.02±3.14
			OA	80.23±0.47	97.96±0.39	95.66±0.57	83.66±0.73
✓	✗	✓	AA	87.14±0.18	98.20±0.51	97.29±0.43	85.83±0.74
			Kappa	77.83±0.50	97.75±0.43	95.30±0.62	80.57±0.84
			OA	78.12±1.30	97.10±0.64	94.92±0.62	80.10±1.61
✓	✓	✗	AA	86.07±0.53	97.56±0.65	96.95±0.38	82.86±1.30
			Kappa	75.52±1.44	96.80±0.71	94.50±0.67	76.36±1.85
			OA	74.48±1.84	97.03±0.59	95.26±0.50	77.26±2.67
✗	✓	✓	AA	83.02±1.00	97.36±0.59	97.13±0.43	81.35±1.56
			Kappa	71.48±2.00	96.73±0.65	94.96±0.54	73.21±3.00
			OA	**80.68±0.27**	**98.37±0.26**	**96.23±0.06**	**84.60±0.47**
✓	✓	✓	AA	**87.35±0.28**	**98.56±0.27**	**97.82±0.09**	**86.63±0.64**
			Kappa	**78.32±0.28**	**98.21±0.29**	**95.91±0.07**	**81.66±0.55**

module, especially on IP, with an increase of nearly 6.83%.

2) The proposed SSCA is much better than simple adaptive fusion, indicating that HSIs are enhanced by spatial–spectral data and then transmitted to the SSCA. The SSCA can better achieve adaptive fusion of local spatial features and global spectral dependencies. From the experimental results, it can be observed that the accuracy of the four datasets will be improved after introducing this module. It is particularly important that the combination of the SSCA and SSCL has significant improvements in all four datasets, especially on IP, with an increase of nearly 7.46%. This indicates that the SSCA can best fuse the spatial–spectral information enhanced by SSCL.
3) The proposed PMC is much better than simple partial convolution (PConv), as it can better extract spatial–spectral information of HSIs. From the experimental results, it can be observed that the accuracy of the four datasets introduced with this module will be slightly improved. It is particularly important to note that the combination of PMC and SSCL has a significant improvement in all four datasets, with an average improvement of nearly 5.30%. Especially in the IP and UP datasets, it has improved by 9.57% and 9.23%, respectively. This indicates that the PMC can better interact with the SSCL, extract the enhanced information of the SSCL, and further integrate with the SSCL. BCL interacts to improve the training speed of the model and alleviate overfitting.

TABLE VIII

PERFORMANCE COMPARISON OF DIFFERENT CONVOLUTION METHODS ON IP DATASET

Metrics	PMC	PConv	2DConv
OA/%	**80.68±0.27**	80.18±3.32	79.43±3.28
Params/K	**1.46**	23.4	16.5
FLOPs/M	**10.48**	170.65	119.44
Time/s	**27.56**	27.62	27.64

TABLE IX

ABLATION STUDY OF THE THREE SUBMODULES (MSE, BSE, AND BCL) IN SSCL

Components			Metrics	Datasets			
MSE	BSE	BCL		IP	KSC	Salinas	UP
✗	✗	✗	OA	74.48±1.84	97.03±0.59	95.26±0.50	77.26±2.67
✓	✗	✗	OA	78.48±0.55	97.62±0.61	95.49±0.26	79.09±2.90
✗	✓	✗	OA	78.30±0.85	97.24±0.71	95.35±0.76	78.77±2.02
✗	✗	✓	OA	75.27±1.04	97.10±0.45	95.41±0.57	78.11±2.04
✓	✓	✗	OA	79.28±0.68	97.65±0.63	95.59±0.39	81.52±2.75
✓	✗	✓	OA	80.54±0.48	97.81±0.54	95.55±0.91	82.36±2.27
✗	✓	✓	OA	78.39±1.28	97.29±0.78	95.55±0.91	80.68±1.82
✓	✓	✓	OA	**80.68±0.27**	**98.37±0.26**	**96.23±0.06**	**84.60±0.47**

To evaluate the computational efficiency and robustness improvements of PMC, comparative experiments were conducted against 2DConv and PConv on the IP dataset using four metrics: OA, Params (K), FLOPs (M), and Time. As shown in Table VIII, superior performance of PMC was observed across all metrics. Compared with PConv, a significant reduction in FLOPs (160.17 M decrease) was achieved, along with moderate improvements in OA (0.50% increase), parameter reduction (21.94 K decrease), and faster inference (0.06 s acceleration). When compared to 2DConv, more noticeable OA enhancement (1.25% improvement) with reduced variance (0.55 decrease) was demonstrated. To analyze the roles of MSE, BSE, and BCL within SSCL, we further compared different combinations of all submodules. As shown in Table IX: 1) MSE can simultaneously extract multigranularity spatial features, and its performance improvement is more pronounced than that of BSE and BCL; and 2) in contrast, the combination of MSE for spatial feature extraction and BCL for pulling together same-class samples while pushing apart different-class samples effectively addresses both overfitting and feature insufficiency in few-shot classification, demonstrating significantly better results than other combinations.

D. Effects of Labeled Training Sample Numbers

To investigate the classification performance of our proposed model under different training samples, we randomly selected 1, 3, 5, 10, 15, and 20 samples from each category in the IP, KSC, Salinas, and UP datasets for comparative experiments. As shown in Fig. 10, as the number of samples increases, the classification performance of all methods gradually improves, indicating that the number of labeled samples

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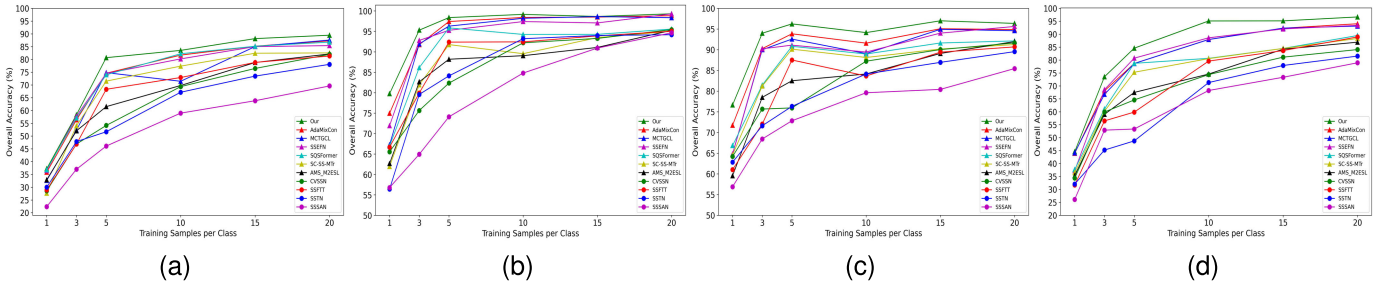


Fig. 10. Evaluation of OA as a function of the number of labeled samples per class. (a) IP. (b) KSC. (c) Salinas. (d) UP.

TABLE X
COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY IN IP DATASETS

Metric	Method										
	SSSAN	SSTN	SSFTT	CVSSN	AMS_M2ESL	SC-SS-MTr	SQSFormer	SSEFN	MCTGCL	AadMixCon	SSECLN
Params/K	313.91	240.95	252.81	260.91	3760.23	722.62	1318.13	16353.68	897.36	1250.54	1120.31
FLOPs/M	672.88	55.87	234.42	337.23	322.06	7273.57	3315.63	5935.85	11952.23	9060.96	1163.32
Traning time/s	26.31	17.49	26.46	10.58	29.35	192.57	32.29	33.22	235.93	186.13	27.56
Test time/s	31.66	2.91	2.85	1.52	1.17	4.91	7.53	8.03	1.46	24.42	1.09

plays a crucial role in training these deep learning models. Furthermore, with the same number of training samples, our method consistently outperforms the other few labeled models in terms of OA (SQSFormer [25], SSEFN [24], MCTGCL [26], AdaMixCon [43]).

E. Computational Complexity Analysis

As presented in Table X, the computational complexity of all methods is evaluated on the IP dataset in terms of the parameter count, FLOPs, training time, and inference time. In the method of few-shot comparison, SSECLN boasts the lowest number of FLOPs (1163.32 M) and the fewest parameters (1120.31 K), indicating its relatively lightweight architecture. In terms of efficiency, SSECLN requires only 27.56 s for training and 1.09 s for inference, both of which are lower than those of advanced few-shot-based methods.

Compared with state-of-the-art few-shot methods, SSECLN employs PMC to reduce FLOPs and model parameters. During training, it utilizes lightweight modules for efficient feature extraction, integrates CL optimization to reduce computational overhead, and specifically enhances spatial–spectral feature extraction, surpassing existing methods in few-shot classification performance.

TABLE XI
NUMBER OF SOURCE AND TARGET SAMPLES FOR PAVIA AND HOUSTON DATASETS

Dataset	Number of Samples by Class							Total
	Tree	Asphalt	Brick	Bitumen	Shadow	Meadow	Bare Soil	
UP	3064	6631	3682	1330	947	18649	5029	**39332**
PC	7598	9248	2685	7287	2863	3090	6584	**39355**
Total	10662	15879	6367	8617	3810	21739	11613	**78687**
	Grass-h	Grass-s	Trees	Water	Res-b	N-res b	Road	
HT13	345	365	365	285	319	408	443	**2530**
HT18	1353	4888	2766	22	5347	32459	6365	**53200**
Total	1698	5253	3131	307	5666	32867	6808	**55730**

TABLE XII
HSI CLASSIFICATION RESULTS ON PAVIA AND HOUSTON DATASETS UNDER DIFFERENT TRAIN-TEST CONDITIONS

Train	Test	OA (%)				
		SQSFormer	SSEFN	MCTGCL	AdaMixCon	SSECLN
PC	PC	86.42±3.70	86.60±1.35	88.02±2.77	90.45±1.03	**94.40±0.29**
PC	UP	49.52±7.71	54.65±6.51	57.35±6.23	65.25±6.15	**75.12±5.15**
HT13	HT13	70.53±6.12	74.53±6.23	78.30±3.33	81.63±2.25	**85.06±2.10**
HT13	HT18	48.76±8.53	52.73±7.56	58.53±6.52	65.13±4.20	**76.34±2.16**

F. Performance on Cross-Dataset HSI Classification

To verify the generalization performance of the proposed method, we conducted cross-scenario experiments on Pavia dataset. This dataset consists of four subsets: Pavia Center (PC), University of Pavia (UP), Houston 2013 (HT13), and Houston 2018 (HT18), with the corresponding data for each category shown in Table XI. PC has a spatial resolution of 1096×715 pixels and 102 spectral bands, while UP has a spatial resolution of 610×340 pixels and 103 spectral bands. To maintain spectral consistency, UP removed the last band. HT13 has a spatial resolution of 349×1905 pixels and 144 spectral bands. The HT18 dataset has the same wavelength range but contains 48 spectral bands. To maintain spectral consistency, HT13 extracted 48 spectral bands (wavelength range 0.38~1.05 μm). These four datasets all have seven categories.

The experiment used the PC (HT13) dataset to train the model and tested it on both the PC (HT13) and UP (HT18) datasets. As shown in Table XII, when the test set changed from PC (HT13) to UP (HT18), the classification performance of all models showed a significant decrease. Specifically, the SQSFormer method achieved a classification accuracy of

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86.42% on the PC test set, while it was 49.52% on the UP test set. In contrast, the SSECLN method demonstrates superior cross-domain performance, with significantly higher classification accuracy than existing methods such as SQSFormer, SSEFN, MCTGCL, and AdaMixCon. This phenomenon is mainly due to the domain shift problem between the training set and the test set.

V. Conclusion

In this study, an innovative SSECLN is proposed to address the key challenges in few-shot HSI classification. This network first utilizes MSE and bidirectional spectral enhancement mechanisms, using parallel convolution paths to extract multiscale spatial features and establish bidirectional spectral modeling, significantly improving the comprehensiveness of feature expression. On this basis, two parallel modules, SSCA and PMC, were innovatively combined. The former achieved dynamic interaction and adaptive fusion of spatial–spectral dimensions, while the latter significantly improved computational efficiency by selectively processing key channels; the final introduced BCL effectively alleviates the overfitting problem while enhancing feature discriminability by optimizing the distribution of features within the batch. Experimental verification on four benchmark datasets shows that the SSECLN significantly outperforms existing methods in classification performance while maintaining high computational efficiency. Future research will focus on developing more lightweight model architectures while maintaining classification performance.

References

[1] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral–spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.

[2] B. Du, Y. Zhang, L. Zhang, and D. Tao, "Beyond the sparsity-based target detector: A hybrid sparsity and statistics-based detector for hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5345–5357, Nov. 2016.

[3] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[4] N. Zhang et al., "Progress and prospects of hyperspectral remote sensing technology for crop diseases and pests," *Nat. remote Sens. Bull.*, vol. 25, no. 1, pp. 403–422, 2021.

[5] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[6] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[7] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[8] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[9] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[10] B. Zhang, Y. Chen, S. Xiong, and X. Lu, "Hyperspectral image classification via cascaded spatial cross-attention network," *IEEE Trans. Image Process.*, vol. 34, pp. 899–913, 2025.

[11] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[12] K. Xu, Z. Zhu, B. Wu, and C. Fan, "Uncertainty-informed prototype contrastive learning for cross-scene hyperspectral image classification," *Knowledge-Based Syst.*, vol. 311, Feb. 2025, Art. no. 113165.

[13] L. Chen, J. Wu, Y. Xie, E. Chen, and X. Zhang, "Discriminative feature constraints via supervised contrastive learning for few-shot forest tree species classification using airborne hyperspectral images," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113710.

[14] R. Wu, H. Liu, Z. Yue, C.-W. Sham, and J.-B. Li, "Feature space expansion and compression with spatial–spectral augmentation for hyperspectral image class-incremental learning," *Pattern Recognit.*, vol. 168, Dec. 2025, Art. no. 111830.

[15] M. Zhu, J. Fan, Q. Yang, and T. Chen, "SC-EADNet: A self-supervised contrastive efficient asymmetric dilated network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519517, doi: 10.1109/TGRS.2021.3131152.

[16] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[17] K.-K. Huang, C.-X. Ren, H. Liu, Z.-R. Lai, Y.-F. Yu, and D.-Q. Dai, "Hyperspectral image classification via discriminative convolutional neural network with an improved triplet loss," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107744.

[18] K. Yang, H. Sun, C. Zou, and X. Lu, "Cross-attention spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[19] D. Peng et al., "Robust hyperspectral image classification using a multiscale transformer with long- and short-distance spatial–spectral cross attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535014, doi: 10.1109/TGRS.2024.3474736.

[20] J. Li, Z. Zhang, R. Song, H. Xu, Y. Li, and Q. Du, "Contrastive MLP network based on adjacent coordinates for cross-domain zero-shot hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 8, pp. 8377–8390, Aug. 2025.

[21] Y. Zhou, X. Huang, X. Yang, J. Peng, Y. Ban, and N. Jiang, "MSMT-LCL: Multiscale spatial–spectral masked transformer with local contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534316, doi: 10.1109/TGRS.2024.3472066.

[22] W. Wei, L. Tong, B. Guo, J. Zhou, and C. Xiao, "Few-shot hyperspectral image classification using relational generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539016, doi: 10.1109/TGRS.2024.3490537.

[23] M. Zhang, Z. Wang, X. Wang, M. Gong, Y. Wu, and H. Li, "Features kept generative adversarial network data augmentation strategy for hyperspectral image classification," *Pattern Recognit.*, vol. 142, Oct. 2023, Art. no. 109701.

[24] S. Liu, C. Fu, Y. Duan, X. Wang, and F. Luo, "Spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502414.

[25] N. Chen, L. Fang, Y. Xia, S. Xia, H. Liu, and J. Yue, "Spectral query spatial: Revisiting the role of center pixel in transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402714, doi: 10.1109/TGRS.2024.3361652.

[26] B. Xi et al., "MCTGCL: Mixed CNN-transformer for Mars hyperspectral image classification with graph contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503214, doi: 10.1109/TGRS.2025.3529996.

[27] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718, doi: 10.1109/TGRS.2023.3264235.

[28] J. Tian et al., "A multimodal prototype correction and multidimensional knowledge distillation framework for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5522718, doi: 10.1109/TGRS.2025.3601337.

[29] L. Shen et al., "Is meta-learning effective for few-shot hyperspectral image classification?," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5531114.

[30] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[31] J. Li, X. Li, and Y. Yan, "Unlocking the potential of data augmentation in contrastive learning for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 12, p. 3123, Jun. 2023.

[32] Y. Zhang, L. Liang, J. Mao, Y. Wang, and L. Jia, "From global to local: A dual-branch structural feature extraction method for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1778–1791, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:48:13 UTC from IEEE Xplore. Restrictions apply.

[33] Y. Zhang, P. Duan, L. Liang, X. Kang, J. Li, and A. Plaza, "PFS3F: Probabilistic fusion of superpixel-wise and semantic-aware structural features for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 9, pp. 8723–8737, Sep. 2025.
[34] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10823–10837, Dec. 2023.
[35] Y. Chen, W. Su, J. Li, and Z. Sun, "Hierarchical object oriented classification using very high resolution imagery and LiDAR data over urban areas," *Adv. Space Res.*, vol. 43, no. 7, pp. 1101–1110, Apr. 2009.
[36] Z. Zhao, J. Song, D. Zhao, J. Zhang, H. Zhou, and J. Zhou, "Multi-scale frequency-guided two-stream network for hyperspectral anomaly detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104355.
[37] Y. Cheng, W. Zhang, H. Wang, and X. Wang, "Causal meta-transfer learning for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521014, doi: 10.1109/TGRS.2023.3309055.
[38] X. Zhang et al., "Spectral–spatial self-attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512115, doi: 10.1109/TGRS.2021.3102143.
[39] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715, doi: 10.1109/TGRS.2021.3115699.
[40] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.
[41] M. Li, Y. Liu, G. Xue, Y. Huang, and G. Yang, "Exploring the relationship between center and neighborhoods: Central vector oriented self-similarity network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1979–1993, Apr. 2023.
[42] M. Li, W. Li, Y. Liu, Y. Huang, and G. Yang, "Adaptive mask sampling and manifold to Euclidean subspace learning with distance covariance representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508518.
[43] C. Wang, J. Sheng, J. Zhou, Z. Song, J. Fan, and Z. Lin, "AdaMixCon: Difficulty-aware feature enhancement for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5504215.



Shizhe Zhang received the B.S. degree in data science and big data technology from Shihezi University, Shihezi, China, in 2023. He is currently pursuing the M.S. degree in artificial intelligence with Guangxi University, Nanning, China.

His research interests include machine learning and image processing.



Lina Yang (Member, IEEE) received the B.S. degree in computer engineering from Shijiazhuang Railway University, Shijiazhuang, China, in 2005, the M.Eng. degree in computer science from the University of Malaya, Kuala Lumpur, Malaysia, in 2011, and the Ph.D. degree from the University of Macau, Macau, China, in 2015.

She is currently an Associate Professor with the School of Computer, Electronics, and Information, Guangxi University, Nanning, China. Her research interests include medical biometrics, machine learning, image processing, and information security.



Yaojun Cai received the master's degree in cartography and geographic information systems from Nanning Normal University, Nanning, China, in 2015.

He is currently the Deputy Director of the Big Data Development Research Institute, Information Center of Guangxi Zhuang Autonomous Region. His research interests include machine learning, image processing, and big data development research.



Yifeng Tan (Student Member, IEEE) received the B.S. degree in computer science and technology from Guangxi University for Nationalities, Chongzuo, Guangxi, China, in 2020, and the Ph.D. degree in computer science and technology from Guangxi University, Nanning, Guangxi, in 2026.

He is currently engaged in research on multimodal learning and hyperspectral image processing.



Wanxing Zhu received the B.S. degree in software engineering from East China University of Technology, Nanchang, China, in 2023. He is currently pursuing the M.S. degree with Guangxi Key Laboratory of Multimedia Communications and Network Technology, Guangxi University, Nanning, China.

His research interests include artificial intelligence, image processing, and computer vision.



Thomas Xinzhang Wu (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the University of Pennsylvania, Philadelphia, PA, USA, in 1999.

Then, he joined the University of Central Florida, Orlando, FL, USA, as an Assistant Professor in 1999, and was promoted to an Associate Professor in 2005 and a Professor in 2011. He also got his tenure in 2005. He is currently a Distinguished Professor with the School of Electrical Engineering, Shanghai Jiaotong University, Shanghai, China. His research interests include machine learning, high-performance motors, and intelligent equipment.



Xichun Li received the B.S. degree in computer science from Northeastern University, Shenyang, China, in 1997, and the M.S. and Ph.D. degrees in computer science from the University of Malaya, Kuala Lumpur, Malaysia, in 2002 and 2010, respectively.

He is currently a Professor of Guangxi Normal University for Nationalities, Chongzuo, Guangxi, China. His research interests include mobile computing.



Yan Yuan Tang (Life Fellow, IEEE) is currently the Chair Professor of the Faculty of Science and Technology, University of Macau, Macau, China, and a Professor/an Adjunct Professor/an Honorary Professor at several institutes, including Chongqing University, Chongqing, China, Concordia University, Montreal, QC, Canada, and Hong Kong Baptist University, Hong Kong, China. His current interests include wavelets, pattern recognition, and image processing. He has published more than 400 academic articles and is the author/co-author of more than 25 monographs/books/book chapters.

Dr. Tang is an IAPR Fellow. He is the Founder and the Editor-in-Chief of *International Journal of Wavelets, Multiresolution and Information Processing* (IJWMIP), and an associate editor of several international journals. He is the Founder and the Chair of the Pattern Recognition Committee in IEEE SMC. He served as the general chair, the program chair, and a committee member for many international conferences. He is the Founder and the General Chair of the Series International Conferences on Wavelets Analysis and Pattern Recognition (ICWAPRs). He is the Founder and the Chair of Macau Branch of the International Association of Pattern Recognition (IAPR).

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